ENTERPRISE VAT MANAGEMENT SYSTEM

Angular Frontend - Complete Professional Implementation Plan

Microservices-integrated frontend blueprint with screen wireframes, detailed modules, dynamic menu, route strategy, roles, approval workflows, validation, accessibility, testing and delivery roadmap.

Scope: Manufacturing | Trading | Services | Multi-company | Multi-branch | Bangladesh VAT compliance

Edition: October 2026 | Document type: Solution architecture and UI specification

Deliverable	Coverage
Screen catalog	70+ proposed screens across 13 domains
Visual wireframes	Dashboard, list, transaction entry, approvals and return preparation
Architecture	Angular standalone features + API gateway + 12 microservices
Governance	Dynamic navigation, RBAC, company and branch scope, approval segregation
Engineering	Routes, components, contracts, validation, test strategy, roadmap

Important: All screen layouts are proposed designs, not screenshots of Shampan VAT. VAT form fields, calculation rules and filing procedures require validation against currently applicable NBR rules.

1. Executive frontend architecture

Build one Angular enterprise shell with feature-based lazy loading, independently owned domain modules and a shared design system. Backend microservices are accessed only through the API Gateway; frontend code does not connect directly to PostgreSQL or RabbitMQ.

Concern	Design choice
Angular	Current supported Angular release at implementation; standalone components, signals, reactive forms
UI / Styling	PrimeNG or Angular Material (select one), design tokens, SCSS
Data access	Typed HttpClient clients, interceptors, RxJS, signal-based view state
Auth	OIDC authorization-code + PKCE, backend token validation, no sensitive tokens in localStorage
Internationalization	English + Bangla, locale-aware date, money and print templates
Reporting	Server-generated PDF/Excel with frontend previews and filters
Deploy	Nginx static build, versioned assets, CSP, environment configuration

Shell composition

App Shell -> Topbar + Sidebar + Company Context + Breadcrumb -> Lazy Feature Route -> Page Container -> Data Grid / Form / Drawer / Dialog -> API Gateway.

Frontend quality targets

- Desktop-first 1440px layouts, responsive tablet and essential mobile read-only screens.
- Keyboard accessible workflows, visible focus, color contrast and screen-reader labels.
- No hidden authorization assumptions: route guard is UX only; every API enforces permission.
- All data grids use server-side pagination and explicit export limits.

2. Information architecture and complete menu

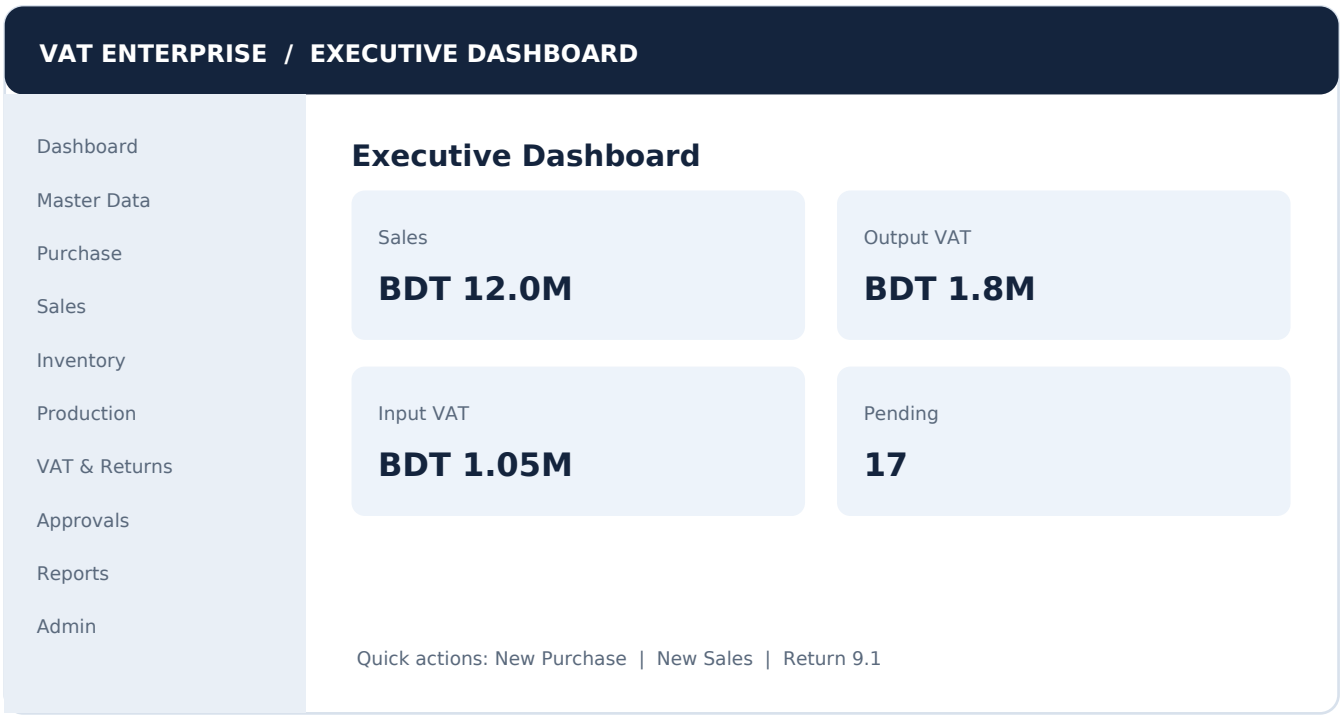
Top-level menu	Submenus / capability
Dashboard	Executive overview, VAT KPI, alerts, branch comparison, pending tasks
Administration	Company, BIN, branches, fiscal years, warehouses, tax periods, number sequences
Security	Users, roles, permissions, role-menu mapping, approval matrix, audit
Master Data	Customer, supplier, item, HS code, UOM, tax code, tax rule, currencies
Purchase & Import	PO, GRN, local invoice, import BOE, return, input credit, 6.1
Sales & Services	Sales order, invoice, 6.3 challan, service invoice, returns, notes, 6.2
Manufacturing	BOM, coefficient 4.3, work order, issue, production receipt, scrap
Inventory	Receipt, issue, transfer 6.5, adjustment, stock ledger, reconciliation
VAT Engine	Tax preview, input/output ledger, VDS 6.6, adjustments, exceptions
Compliance	Mushak forms, return 9.1, reconciliation, filing record, period lock
Approvals	My tasks, checker queue, approver queue, returned, delegation history
Reports & MIS	Liability, transaction registers, reconciliation, aging, audit, exports
Integrations	ERP jobs, file upload, mapping, retry queue, sync logs

3. Detailed screen inventory

Domain	Screens	Count
Dashboard	Executive, VAT KPI, Sales/Purchase Trends, Alerts, My Tasks	5
Administration	Company List/Edit, BIN Profile, Branches, Periods, Warehouses, Sequences	6
Security	Users, User Editor, Roles, Role Matrix, Menu Builder, Approval Matrix, Audit	7
Master	Customers, Suppliers, Items, HS Codes, UOM, Tax Rules, Tax Rate History	7
Purchase	PO List/Edit, GRN, Invoice List/Edit, Import BOE, Returns, 6.1	7
Sales	Order, Invoice List/Edit, 6.3 Preview, Services, Return, Credit, Debit, 6.2	8
Manufacturing	BOM, 4.3, Production Order, Material Issue, Receipt, Wastage	6
Inventory	Stock Summary, Movement, Transfer, Adjustment, Count, Reconcile	6
VAT	Tax Preview, Ledger, VDS, Adjustment, Exceptions, Rule Simulation	6
Compliance	6.1/6.2, 6.3, 6.5/6.6, 9.1 Draft, Reconcile, Filing, Lock	7
Workflow	My Tasks, Checker, Approver, Return, Rejection, History	6
Reports	VAT Liability, Input Credit, Sales/Purchase, Branch, Audit, Scheduled	6
Integration	Connections, Import Wizard, Sync Jobs, Failures, Webhooks	5

Total proposed screens: 82. The count represents functional page types; individual forms may include tabs, drawers and reusable dialogs.

Executive Dashboard



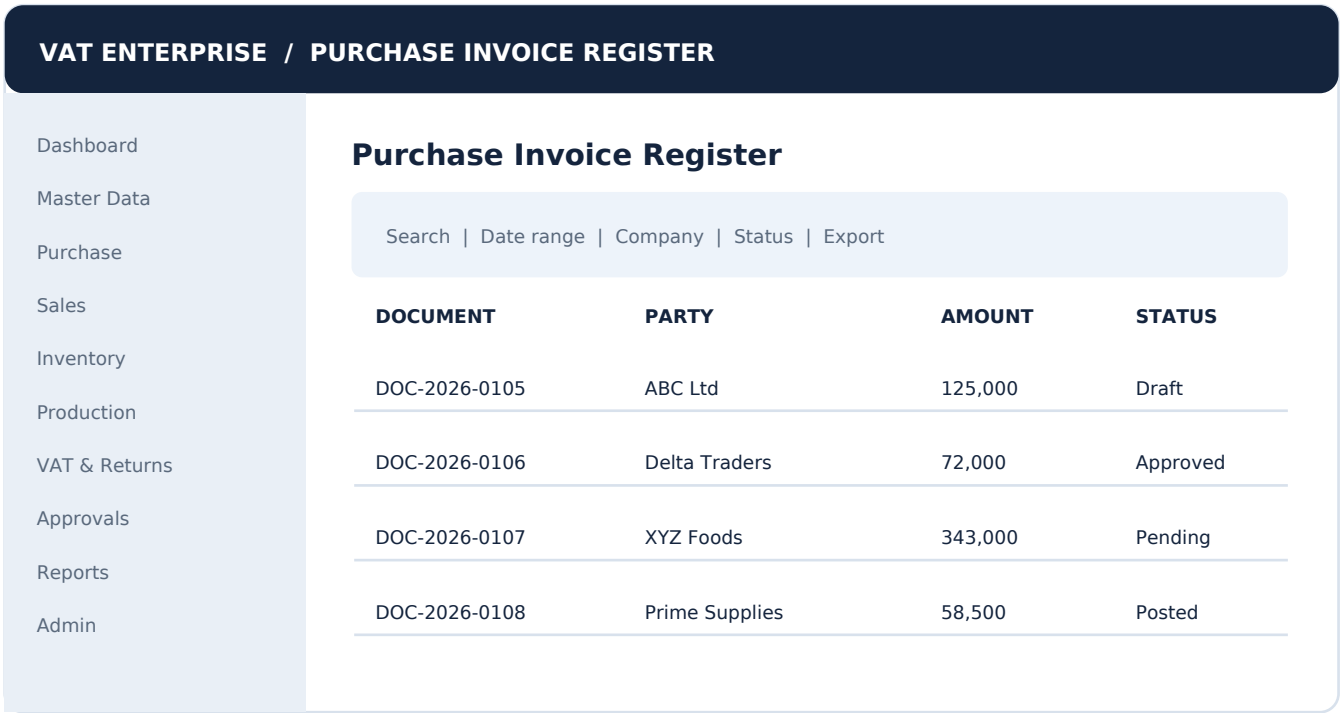
Screen specifications

Fields / filters: Company, branch, reporting month, VAT period, business unit, comparison period.

Actions: Drill into sales, input credit, output VAT, liability, pending approvals, exceptions and return readiness.

Validation / behavior: KPI totals are derived from authorized reporting APIs; distinguish provisional and posted data; show data refresh timestamp.

Purchase Invoice Register



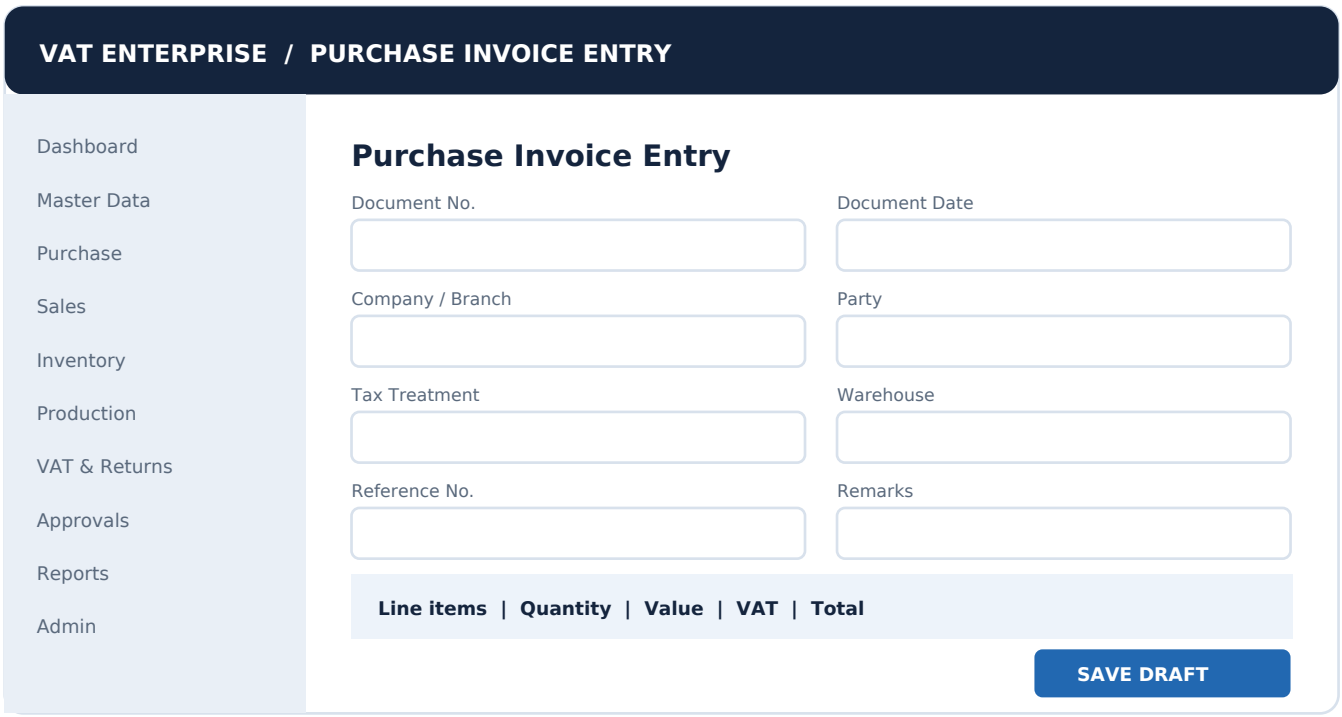
Screen specifications

Fields / filters: Invoice number, supplier, date range, BIN, posting state, VAT status, branch, amount.

Actions: Search, view, new, export, submit, print, bulk-select allowed records.

Validation / behavior: Posted invoices read-only; permission-gated actions; export respects tenant and server filter scope.

Purchase Invoice Entry



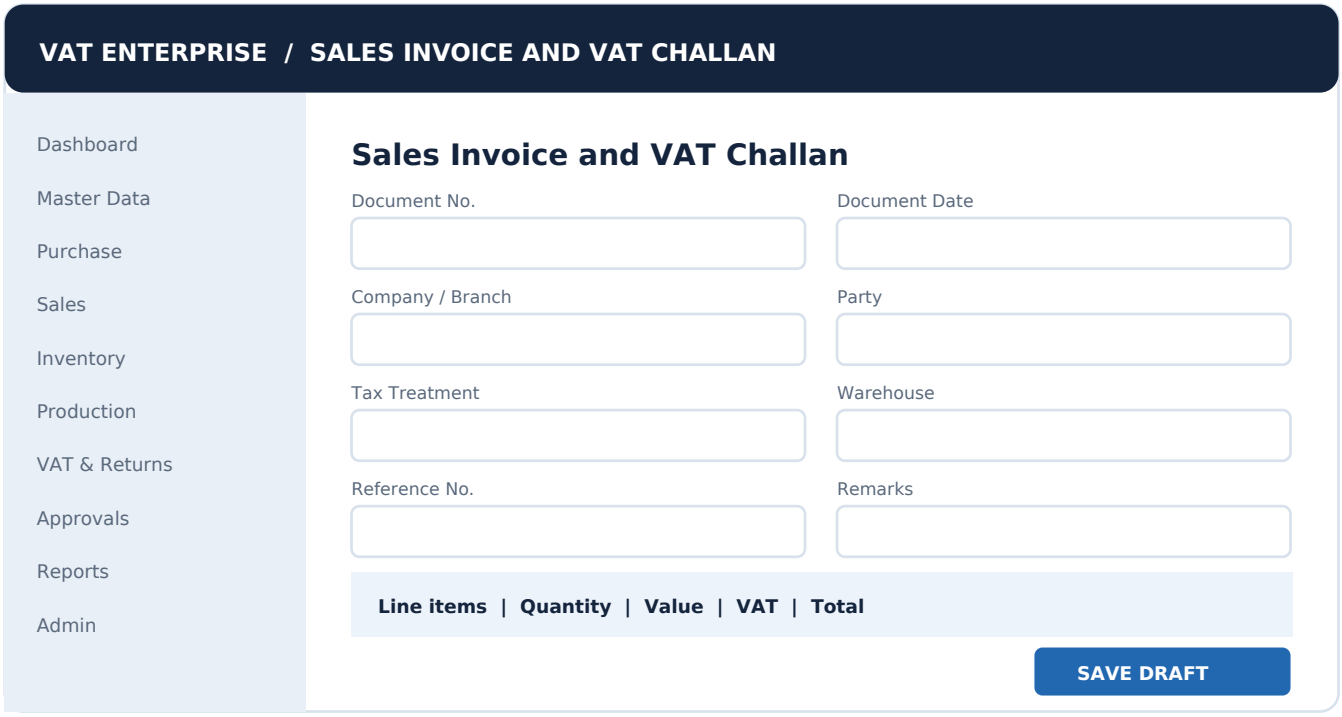
Screen specifications

Fields / filters: Supplier/BIN, source PO/GRN, invoice date, tax point, item lines, taxable value, tax code, VAT amount, supporting documents.

Actions: Save draft, add line, tax preview, attach, submit, return to list.

Validation / behavior: Supplier and product selection via master lookup; no client-only tax authority; server-calculated tax preview required before submit.

Sales Invoice and VAT Challan



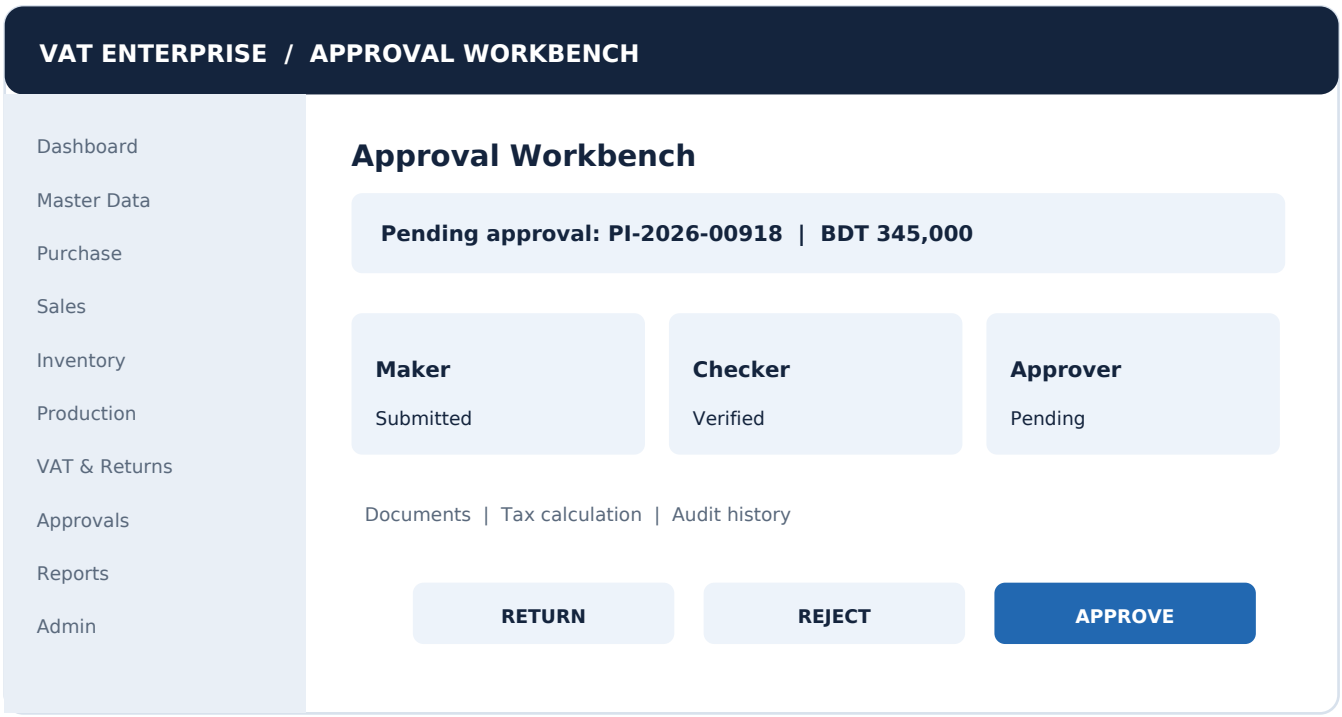
Screen specifications

Fields / filters: Customer/BIN, invoice type, delivery reference, taxable supply lines, VAT treatment, discount, service type, tax amount.

Actions: Save draft, calculate, submit, approve, generate approved challan PDF.

Validation / behavior: Do not issue final tax invoice from unapproved data; preserve document numbering and correction history.

Approval Workbench



Screen specifications

Fields / filters: Company, branch, document type, maker, amount, aging, exception flag, SLA, status.

Actions: Open details, review attachments, return with reason, reject with reason, approve with confirmation.

Validation / behavior: Maker cannot approve own document; backend re-checks role, segregation, current state and optimistic concurrency.

Monthly VAT Return - Mushak 9.1



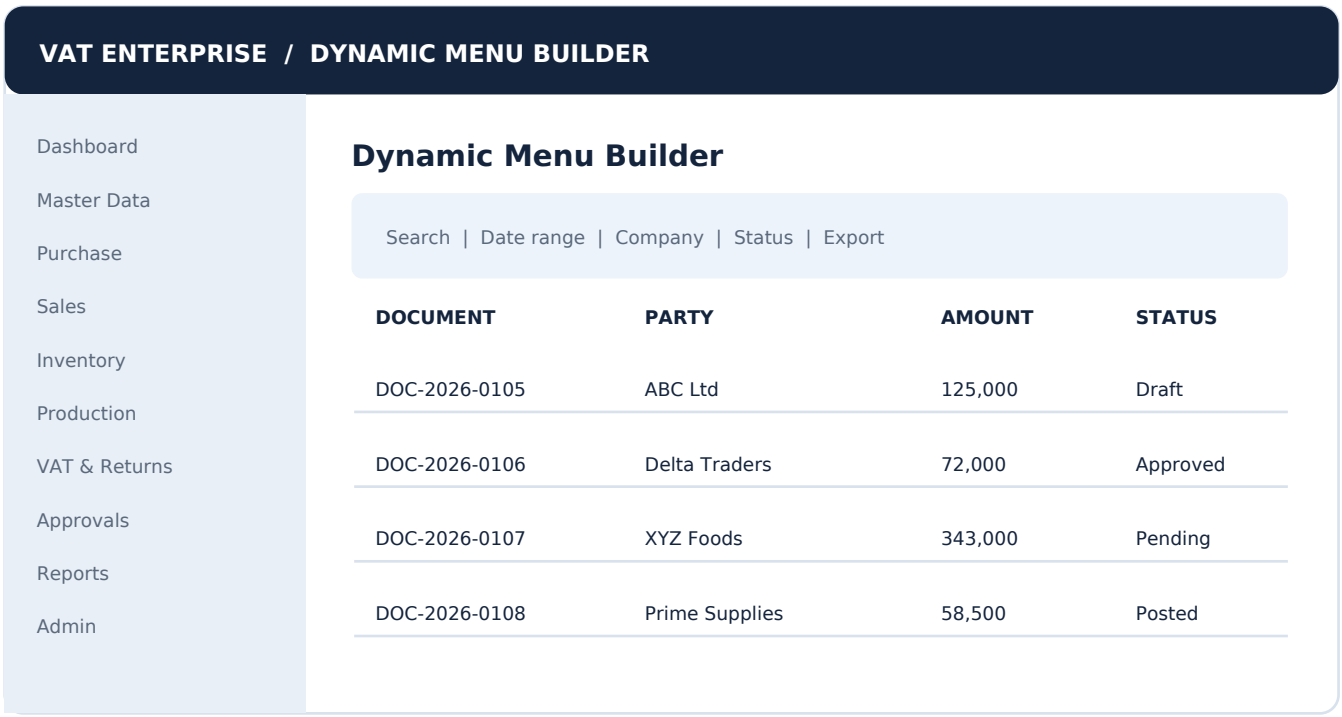
Screen specifications

Fields / filters: Tax period, BIN, filing unit, output tax, eligible input credit, VDS, adjustments, previous balances.

Actions: Generate draft, validate, reconcile, approve, export, record filing acknowledgment, lock period.

Validation / behavior: All values must be traceable to immutable ledger sources; block finalization on unresolved critical exceptions.

Dynamic Menu Builder



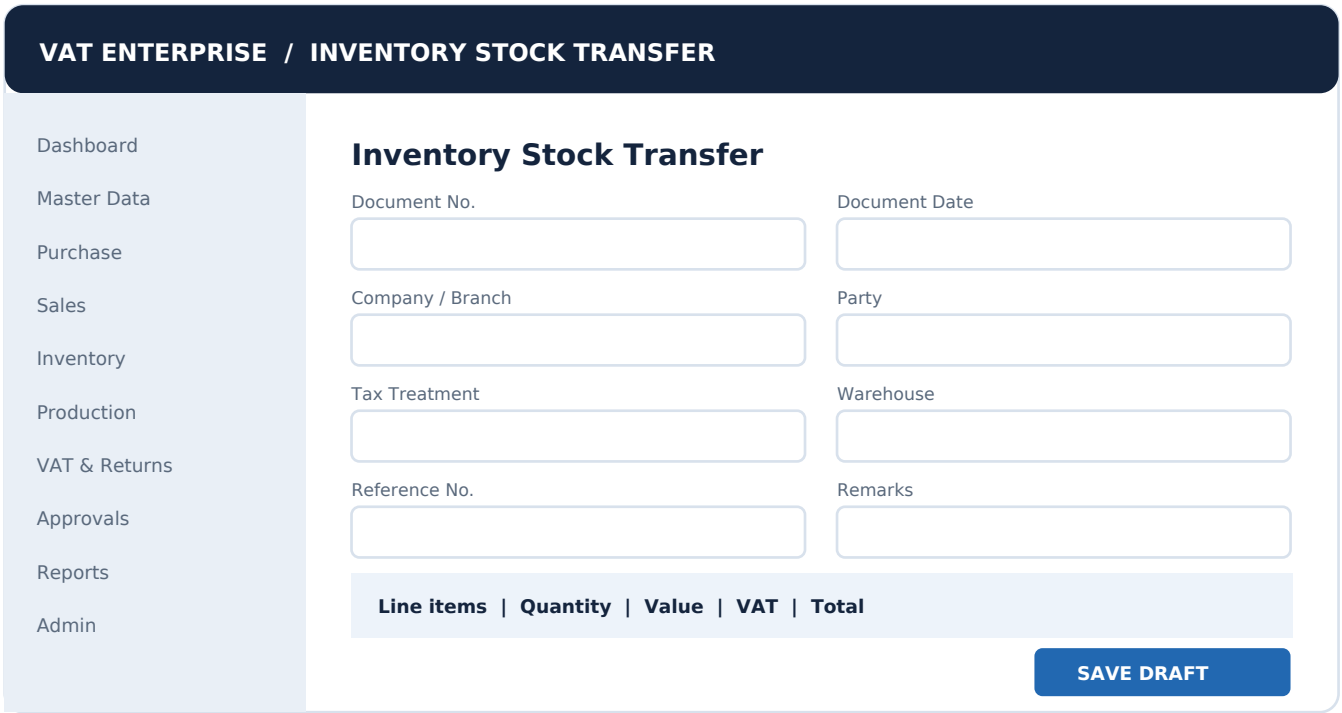
Screen specifications

Fields / filters: Menu code, label EN/BN, icon, route, parent, order, active flag, permission code.

Actions: Create/edit hierarchy, reorder, activate, preview as role, publish menu version.

Validation / behavior: Route allowlist, cycle prevention, duplicate-code checks; server permissions independent of menu visibility.

Inventory Stock Transfer



Screen specifications

Fields / filters: Source branch/warehouse, destination, items, quantity, batch/lot, movement date, transport reference.

Actions: Validate stock, save, submit, approve, dispatch, receive, print transfer document.

Validation / behavior: Prevent negative stock per policy, enforce transfer lifecycle and cross-branch ownership; applicable Mushak output verified.

Manufacturing BOM and Coefficient



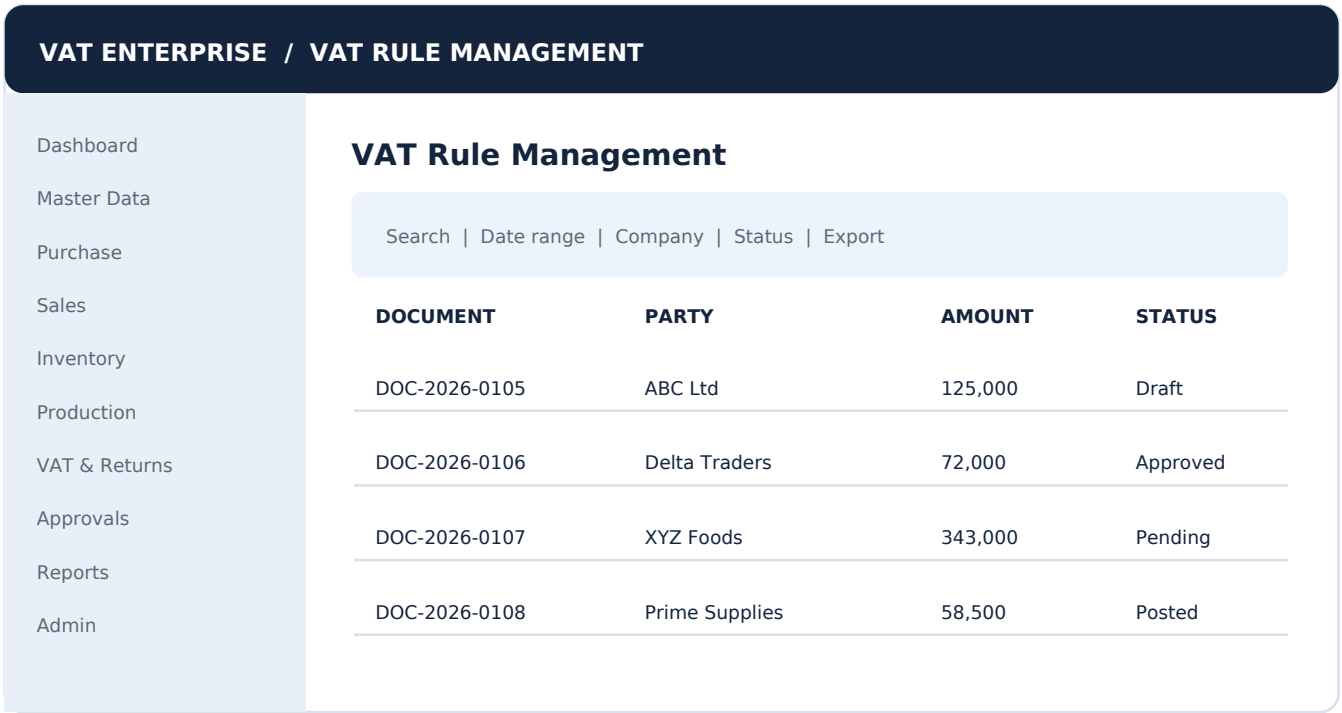
Screen specifications

Fields / filters: Product, BOM revision, raw materials, yield, loss, costing, effective dates, tax classification.

Actions: Draft BOM, calculate coefficient, submit for approval, compare revisions, generate applicable form.

Validation / behavior: Approved revision immutable; no overlapping effective version; trace raw material consumption to production order.

VAT Rule Management



Screen specifications

Fields / filters: Tax code, rule category, supply type, legal reference, rate, effective dates, status.

Actions: Create draft, simulate transaction, submit rule change, compare version, approve and activate.

Validation / behavior: No overlapping active rules in same scope; retroactive changes require controlled migration and audit.

4. Dynamic menu data model and API contracts

The frontend receives a permission-filtered navigation tree after authentication and company selection. Menu nodes are administrative metadata, not an authorization grant.

Menu property	Type / example
id / parentId	number / nullable number
code	VAT_PURCHASE_INVOICE_LIST
labelEn / labelBn	Purchase Invoices / translated label
route	/app/purchases/invoices
icon	receipt
permission	purchase.invoice.read
order / enabled	10 / true
children	Nested menu nodes

Gateway endpoints

Method	Route	Use
GET	/api/v1/navigation/my-menu	Role-filtered menu tree
GET	/api/v1/security/me/permissions	Current effective permissions
GET	/api/v1/companies/accessible	Allowed company contexts
POST	/api/v1/security/menus	Admin create menu
PUT	/api/v1/security/roles/{id}/permissions	Update role permission mapping
GET	/api/v1/workflows/my-tasks	Approval inbox

Cache menu by user + company + permission-version. Invalidate on role, company access or menu publication changes. Never trust a cached menu to authorize a backend action.

5. Routes and guards

Angular route	Feature / permission
/login	OIDC entry and callback
/app/dashboard	dashboard.read
/app/purchases/invoices	purchase.invoice.read
/app/purchases/invoices/new	purchase.invoice.create
/app/sales/invoices/:id	sales.invoice.read
/app/manufacturing/bom	manufacturing.bom.read
/app/inventory/transfers	inventory.transfer.read
/app/vat/ledger	vat.ledger.read
/app/compliance/returns/:id	compliance.return.read
/app/workflow/my-tasks	workflow.task.read
/app/admin/roles	security.role.manage

Route guards: authGuard (session), companyContextGuard (company selected), permissionGuard (UX-level permission), unsavedChangesGuard (dirty forms). API authorization remains the security boundary.

6. Angular project structure

src/app/core/ - auth, API clients, interceptors, error handling, company context, configuration, telemetry.

src/app/layout/ - app shell, topbar, sidebar, breadcrumbs, notifications, company selector.

src/app/shared/ - design system, reusable forms, data grid, file upload, money/date pipes, validators.

src/app/features/ - dashboard, administration, security, master-data, purchase, sales, inventory, manufacturing, vat, compliance, workflow, reporting, integration.

src/app/state/ - authenticated principal, active company, menu tree, locale, shared lookup cache.

src/environments/ - runtime configuration reference; do not ship credentials or secrets.

Recommended component naming

Feature	Components
Purchase	purchase-invoice-list, purchase-invoice-form, purchase-invoice-detail, purchase-line-editor
Sales	sales-invoice-list, sales-invoice-form, vat-challan-preview
VAT	tax-rule-list, tax-calculation-preview, tax-ledger-list
Compliance	return-list, return-editor, return-reconciliation, filing-history
Security	role-list, permission-matrix, menu-tree-editor, user-access-editor
Workflow	my-task-list, approval-detail, decision-dialog, audit-timeline

Reusable UI components

• AppDataTable: server paging, sorting, filtering, column presets, export request.

• DocumentHeaderForm + DocumentLineGrid + TotalsPanel + AttachmentPanel.

• ApprovalTimeline, AuditHistoryDrawer, ValidationSummary, TaxBreakdown.

• MoneyInput, DateRangeFilter, CompanyBranchPicker, ProductLookup, PartyLookup.

• ConfirmDecisionDialog with mandatory reason on return/reject.

7. API integration and frontend state

Each feature owns typed API clients. A gateway URL is configured at runtime. Angular interceptors attach correlation ID and handle 401, 403, 409, 422 and 5xx distinctly. Use cancellation for obsolete searches and debounce filters.

HTTP response	UI behavior
401	Renew through supported auth flow or redirect to login
403	Show access denied; never silently reroute to a privileged page
409	Concurrency conflict; offer reload and comparison
422	Map field errors to reactive controls and summary
429	Respect retry-after; explain temporary rate limit

HTTP response	UI behavior
5xx	Show retryable error with correlation ID

State design: Signals for local view state; RxJS for HTTP and streaming flows. Avoid global stores for every form field. Use shared state only for session, company context, permissions and cross-page reference data.

8. Role and permission matrix

Role	Create	Verify	Approve	Post/Finalize	Admin
Company Admin	Limited	-	-	-	Yes
Purchase Maker	Purchase	-	-	-	-
Sales Maker	Sales	-	-	-	-
Checker	-	Assigned	-	-	-
Approver	-	-	Assigned	By policy	-
VAT Manager	Adjustment	Review	Return	Return	-
Auditor	-	-	-	-	Read-only
System Job	-	-	-	Authorized job	-

Permissions should use verb-based codes, e.g. purchase.invoice.create, purchase.invoice.submit, purchase.invoice.approve, vat.return.finalize, security.menu.manage. Every request must also validate company, branch, document status and segregation-of-duties constraints.

9. Workflow design

Status	Allowed transitions	UI behavior
Draft	Submitted / Cancelled	Editable; show Save and Submit
Submitted	Checked / Returned / Rejected	Maker read-only; checker queue
Checked	Approved / Returned / Rejected	Approver queue with comparison
Approved	Posting / controlled cancellation	Display approved status; no direct edit
Posting	Posted / PostingFailed	Progress indicator and reconciliation reference
Posted	Reversal / adjustment workflow	Immutable detail, print and audit
Returned	Draft after maker correction	Reason displayed prominently
PostingFailed	Retry / manual reconciliation	Support and operations queue

User journey

Maker creates purchase invoice -> saves draft -> runs tax preview -> submits -> Checker reviews source documents -> Approver decides -> Process manager requests inventory and VAT posting -> frontend shows Posted only after completion confirmations -> Reporting projection refreshes.

Failure and recovery

Approval does not equal posting completion. Pending/failed distributed steps must be visible with a correlation ID, retry status and non-destructive correction options.

10. Validation requirements

- Required fields and contextual validation on each document type.
- Taxable amount, tax code, rate and credit eligibility calculated or confirmed by backend.
- Document date cannot fall in locked period.
- Branch, warehouse, supplier and product must be valid for selected company.
- Duplicate external invoice / reference checks; show warnings and blocking errors distinctly.
- File upload size/type checks plus backend malware scan and storage authorization.
- VAT report finalization blocks unresolved critical reconciliation errors.

11. Multi-company and multi-branch UX

Company selector in topbar determines current context; branch selector offers authorized branches only. Switching company clears feature caches, draft state warnings, permissions, menu tree and lookup options. No cross-company leakage in deep links, exports or browser back navigation.

Area	Behavior
Company switch	Confirm unsaved changes, reset caches and reload menu
Branch filter	Constrain options by user access and document scope
Tax period	Show open, closing and locked statuses
Report consolidation	Explicitly identify branch versus company totals
Numbering	Server-owned per company/branch/document series

12. Design system

Element	Specification
Navigation	Collapsible left sidebar 248px; topbar 64px; breadcrumbs
Grid	12-column desktop; 8 tablet; 4 mobile
Spacing	4/8/12/16/24/32px scale
Colors	Navy primary, blue actions, neutral surfaces, semantic error/warning/success
Typography	Readable bilingual typeface with tested Bengali shaping
Tables	Sticky headers, virtualize only when appropriate, keyboard access
Forms	Two-column desktop, single-column mobile, clear required labels
Status	Text + icon + color; never color alone
Print	A4 templates with legal version, BIN, signature/verification placeholders

13. Reporting screens

- VAT executive dashboard with provisional versus posted breakdown.
- Purchase and sales registers with transaction drill-down.
- Input credit eligibility and disallowed credit analysis.
- VDS certificates and withholding reconciliation.
- Mushak 9.1 draft, field-level traceability and filing history.
- Tax exceptions, late approvals, duplicate documents and mismatches.
- Branch consolidation and period-over-period comparisons.
- Audit reports by user, document, action, timestamp and correlation ID.

14. Security and accessibility acceptance

- OIDC PKCE, CSRF protection where cookie-based sessions apply, strict CSP and secure headers.
- No role/permission bypass via query parameters or modified Angular state.
- Server enforces authorization and tenant scope on every read/write.
- WCAG 2.2 AA target: keyboard navigation, focus management, labels and error summaries.
- Audit user, timestamp, old/new values, reason and document reference for sensitive changes.
- Protect confidential tax data in browser caching, logs, screenshots and exports.

15. Test plan

Test suite	Critical scenarios
Unit	Form validators, tax display, route builders, permission utilities
Component	Grid filters, line editors, approval dialog, error messages
Integration	API contract, pagination, token renewal, 409 conflict, upload
E2E	Purchase to approval; sales challan; return reconciliation
Security	Cross-company access, IDOR, role bypass, XSS, CSRF
Accessibility	Keyboard-only, screen reader, focus trap, contrast
Performance	Large register paging, lazy bundle, report loading
UAT	VAT specialist validates current Mushak field mapping

16. Delivery roadmap

Sprint	Frontend deliverables	Exit criteria
1-2	Shell, design tokens, auth, company selector, route guards	Authenticated shell works
3-4	Dynamic menu, users, roles, permission matrix	Permission checks verified
5-6	Master data, customer, supplier, product, tax rules	CRUD and lookups tested
7-9	Purchase, import, sales, invoice and challan	Approved document journey
10-11	Inventory, manufacturing, BOM, coefficient	Stock/production UX tested
12-13	VAT ledger, VDS, adjustments, return screens	Ledger traceability verified
14	Workflow queues, reporting and export	Maker/checker/approver UAT
15-16	Accessibility, hardening, performance, release	Release gate approved

Estimated frontend delivery: 16 two-week sprints if sequentially staffed by a small team; parallel delivery can shorten elapsed time, but NBR mapping and integration may extend the schedule.

17. Definition of done

- Screen matches approved wireframe and responsive breakpoints.
- All action permissions checked in UI and independently on backend.
- Loading, empty, error, conflict, and retry states designed and tested.
- Form validation and audit history available where applicable.
- Bangla/English text, dates, amounts and PDF print reviewed.
- Automated tests pass, accessibility checked and performance budget met.
- VAT specialist signs off compliance-related form mapping.

18. Immediate implementation backlog

Start with: 1) Angular workspace + design system; 2) Auth and context; 3) Dynamic menu API contract; 4) Permission matrix; 5) Purchase invoice list/form; 6) approval inbox; 7) VAT calculation preview; 8) sales challan; 9) return 9.1 draft.

This document is a comprehensive frontend plan and design specification. The embedded screens are original conceptual wireframes, not functioning Angular pages or legally approved Mushak layouts.